

Course: (2026S) COMP-5450-SA - Mobile Programming

Exercise 2: MEMORY MATCHING FLUTTER GAME

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GitHub URL:

Project Overview

This project is a Memory Matching Game developed using Flutter and Dart. The application challenges users to match pairs of shape cards using memory skills.

The game includes:

- Welcome screen
- Interactive memory card gameplay
- Timer functionality
- Move counter
- Pair counter
- Best score tracking
- Restart functionality
- Home navigation

The application was tested successfully on Android Emulator.

Technologies Used

- Flutter
- Dart
- Android Studio
- Android Emulator (Pixel 6)

Features

1. Welcome Screen

- Displays game title
- Start Game button
- User-friendly interface

2. Memory Matching Gameplay

- Shape image cards

- Randomized card arrangement
- Flip and match logic

3. Score Tracking

- Move counter
- Pair counter
- Game timer
- Best score display

4. Navigation

- Home button
- Restart game button

5. Winning Dialog

- Displays moves taken
- Displays time taken
- Play Again option

Project Structure

```
memory_matching_game
├── android
├── ios
├── lib
│   └── main.dart
├── assets
│   └── images
│       ├── red_circle.png
│       ├── blue_square.png
│       ├── green_triangle.png
│       ├── orange_diamond.png
│       ├── purple_star.png
│       └── pink_heart.png
├── pubspec.yaml
└── README.pdf
```

How to Run the Project

Step 1: Install Flutter SDK and Android Studio.

Step 2: Open the project in Android Studio.

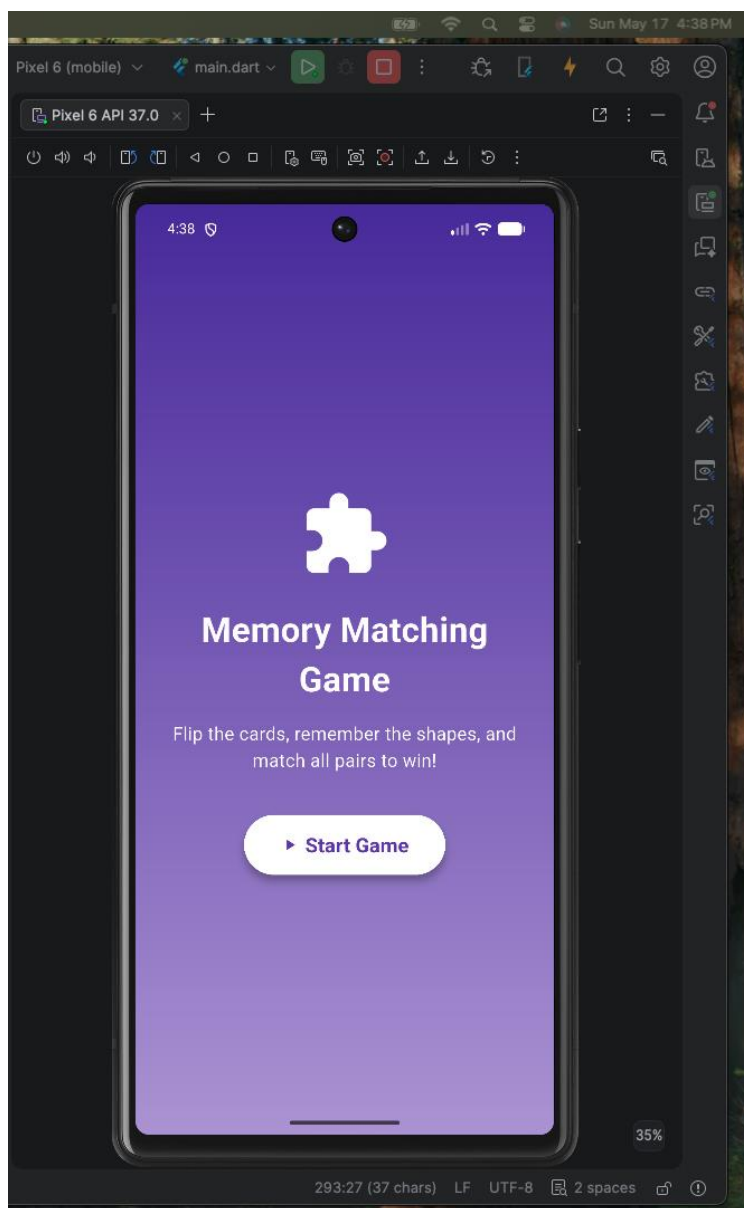
Step 3: Run the following command: `flutter pub get`

Step 4: Start Android Emulator.

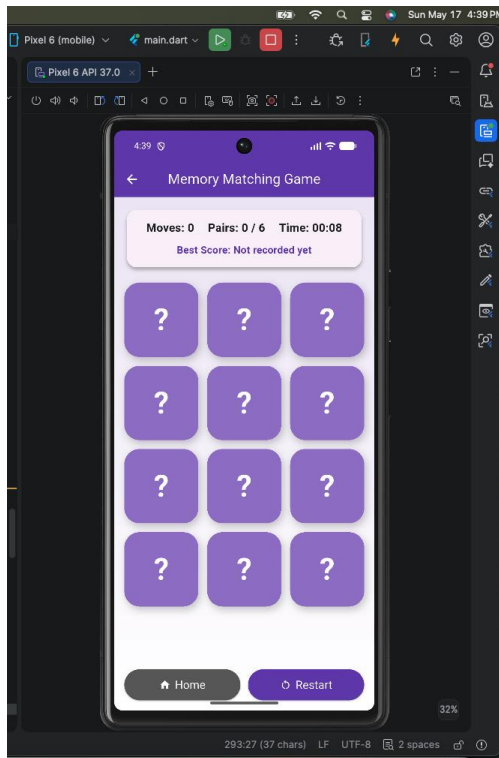
Step 5: Click Run ► in Android Studio

Screenshots

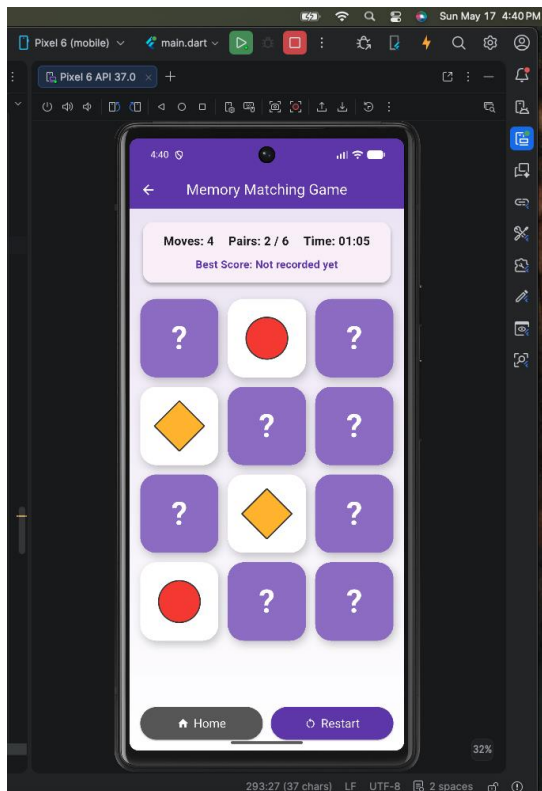
Welcome Screen



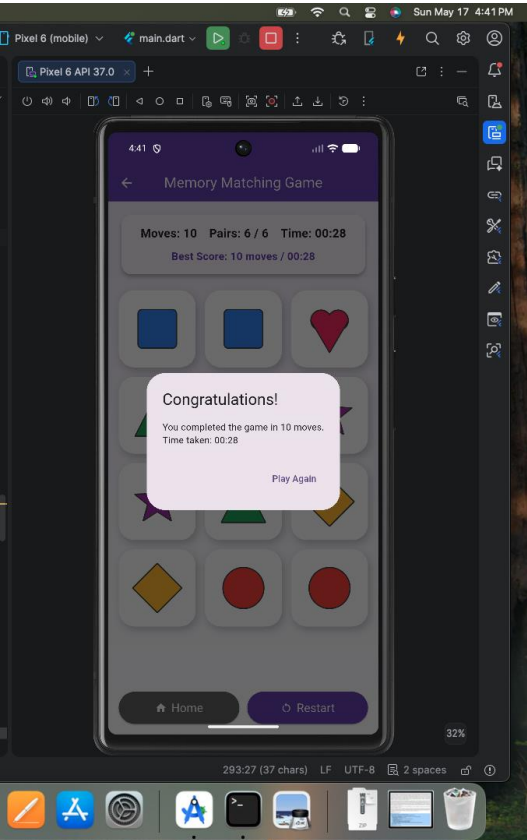
Game Screen



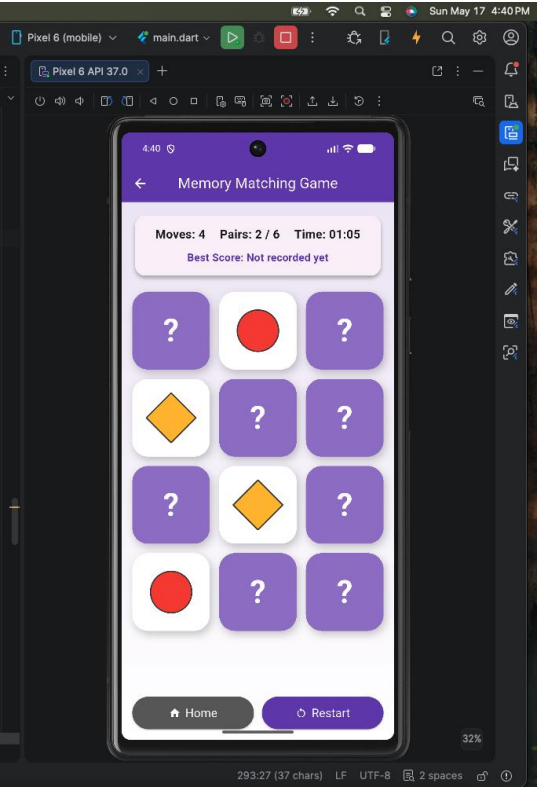
Matching Gameplay



Winning Dialog



Best Score Display



Conclusion

This project demonstrates the implementation of a responsive Memory Matching Game using Flutter and Dart. The application includes interactive gameplay, responsive design, score tracking, timer functionality, and navigation features.